DAA EXPERIMENT 8

Insertion Sort with Duplicate Elements

Aim

To implement Insertion Sort on an array containing duplicate elements.

Algorithm

1. Read the array.
2. Select the element at index 1 as the key.
3. Compare the key with the elements before it.
4. Shift elements greater than the key one position to the right.
5. Insert the key into its correct position.
6. Repeat for all remaining elements and display the sorted array.
7. Stop.

Program

```
# Insertion Sort with Duplicate Elements

arr = [5, 2, 4, 2, 3, 5, 1]

for i in range(1, len(arr)):
    key = arr[i]
    j = i - 1

    while j >= 0 and arr[j] > key:
        arr[j + 1] = arr[j]
        j -= 1

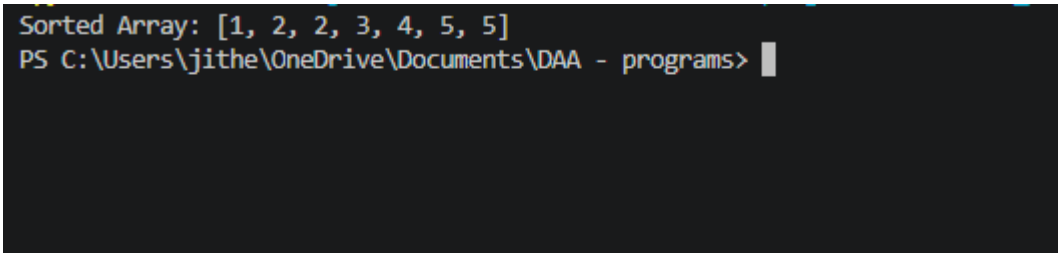
    arr[j + 1] = key

print("Sorted Array:", arr)
```

Complexity Analysis

- Best Case: O(n)
- Average Case: $O(n^2)$
- Worst Case: $O(n^2)$
- Space Complexity: O(1)

Output



Result

Thus, Insertion Sort was successfully implemented on an array containing duplicate elements.